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KERTAS PENERANGAN (INFORMATION SHEET)

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TAHAP / LEVEL	4 (FOUR)	
NO. DAN TAJUK KEBOLEHAN TERAS <i>ICORE ABILITY NO. AND TITLE</i>	CA08 SYSTEM TECHNOLOGY APPLICATION	
NO. DAN PENYATAAN KEBOLEHAN / <i>ABILITY NO. AND STATEMENT</i>	08.01 ADOPT APPROPRIATE SOFTWARE APPLICATION 08.02 COMPLY TO SYSTEM ANALYSIS APPLICATION 08.03 UPDATE IT APPLICATION 08.04 ADHERE TO IT RULES & REGULATIONS	
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TAJUK / TITLE: IT RULES & REGULATIONS ADHERENCE

TUJUAN/PURPOSE :

This Information Sheet is intended to describe on IT Rules and Regulations Adherence requirements to maintain and ensure information technology applications in the organisation are current.

The person who is competent in this CA shall be able to adhere IT rules and regulations in accordance with recommended practices.

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PENERANGAN/ INFORMATION:

1. PRIVACY & SECURITY

1.1. Understanding security and privacy features

Internet Browsers provides a number of features that help to protect the user's privacy and make the computer and the personally identifiable information more secure.

Privacy features allow the user to protect the personally identifiable information by helping the user to understand how Web sites the user view may be using this information and by allowing the user to specify privacy settings that determine whether or not the user wants to allow Web sites to save cookies on the computer.

Privacy features in Internet Browsers include:

- i) Privacy settings that specify how the computer deals with cookies. Cookies are files created by a Web site that store information on the computer, such as the preferences when visiting that site. Cookies may also store personally identifiable information, such as the user name or e-mail address.
- ii) Privacy alerts that let the user know when the user try to gain access to a site that doesn't meet the criteria in the user's privacy settings.
- iii) The ability to view a Web site's P3P privacy policy.

Security features help to prevent people from gaining access to information that the users have not given them permission to access, such as credit card information the user enters when shopping on the Internet. Security features can also protect computer from unsafe software.

Security features in some Internet Browsers include:

- a) Certificates, which verify the identity of a person or the security of a Web site.
- b) 128-bit secure connection for using secure Web sites.
- c) Use security zones to set different levels of security for different areas of the Web to help protect the computer.

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1.2. Securely sharing personal information

There are Internet Browsers that can save the user from having to enter the same information, such as the address or e-mail name, every time the user visit a new Web site that requests such information. It does this by storing the information on the computer. None of this information can be viewed on the computer, or shared with others, without the user's permission.

When a Web site requests information, the request will tell the user:

- i) The Internet address of the site requesting the information.
- ii) What information the site is requesting from Profile Assistant, so the user can choose to exclude information.
- iii) How this information will be used.
- iv) Whether this site has a secure connection (Secure Sockets Layer or SSL). If it does, the user can verify the site's certificate

1.3. Understanding cookies

Internet sites use small files called cookies to keep track of user preferences and information. Internet Browsers include settings that enable the user to control the storage of cookies on the computer.

A cookie is a file created by an Internet site to store information on the user's computer, such as the user's preferences when visiting that site. For example, if the user inquires about a flight schedule at an airline's Web site, the site might create a cookie that contains the user's itinerary. Or it might only contain a record of the pages the user looked at within the site visited, to help the site customize the view for the user's next visit.

Cookies can also store personally identifiable information. Personally identifiable information is information that can be used to identify or contact the user, such as the user's name, e-mail address, home or work address, or telephone number. However, a Web site only has access to the personally identifiable information that the user provide. For example, a Web site cannot determine the user's e-mail name unless the user provides it. Also, a Web site cannot gain access to other information on the computer.

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Once a cookie is saved on the computer, only the Web site that created the cookie can read it.

There are several types of cookies, and the user can choose whether to allow some, none, or all of them to be saved on the computer. If the user does not allow cookies at all, they may not be able to view some Web sites or take advantage of customization features (such as local news and weather, or stock quotes).

i) **Persistent Cookie**

A persistent cookie is one stored as a file on the user's computer, and it remains there when the user close Internet Explorer. The cookie can be read by the Web site that created it when the user visits that site again.

ii) **Session Cookie**

A temporary or session cookie is stored only for the user's current browsing session, and is deleted from the computer when the user closes Internet Explorer.

iii) **First-party cookie**

A first-party cookie either originates on or is sent to the Web site the user is currently viewing. These cookies are commonly used to store information, such as the user's preferences when visiting that site.

iv) **Third-party cookie**

A third-party cookie either originates on or is sent to a Web site different from the one the user is currently viewing. Third-party Web sites usually provide some content on the Web site the user are viewing. For example, many sites use advertising from third-party Web sites and those third-party Web sites may use cookies. A common use for this type of cookie is to track the user's Web page use for advertising or other marketing purposes. Third-party cookies can either be persistent or temporary.

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v) **Unsatisfactory cookies**

Unsatisfactory cookies are cookies that might allow access to personally identifiable information that could be used for a secondary purpose without the user's consent.

1.4. Protecting the computer from unsafe software

When downloading or running programs from the Internet, the user want to know that the program comes from a known, reliable source. That's why, when the user chooses to download a program from the Internet to their computer, the Internet Browser uses certain technology to verify the identity of the program. Authenticode technology verifies that the program has a valid certificate: that the identity of the software publisher matches the certificate, and that the certificate is still valid. Note that this does not prevent a poorly written program from being downloaded or run on the computer, but it does reduce the chance of someone misrepresenting a program that is intended to be malicious or intentionally harmful.

The user can specify different settings for the way Internet Browser handles downloading programs and files, depending on the zone they are coming from.

For example, the user might be confident that anything the user downloads within their corporate intranet is safe. So, the user might set the user security settings for their Local intranet zone to a low level to allow downloading with little or no prompting. If the source is in the 'Internet Zone' or the 'Restricted Sites Zone', the user might want the security levels set to Medium or High. Then, they will be prompted with information about the program's certificate before it is downloaded, or the user might not be able to download it all.

1.5. Using Content Advisor to control access

The Internet provides unique access to a wide variety of information. Some information, however, may not be suitable for every viewer. For example, the user might want to prevent children from seeing Web sites that contain violent or adult content.

Content Advisor provides a way to help control the types of content that the computer can gain access to on the Internet. After turning on Content Advisor, only

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rated content that meets or exceeds criteria can be viewed. The settings can be adjusted.

With Content Advisor, the user can:

- i) Control access to Content Advisor settings. The user turns Content Advisor on, and views the settings, through the use of a password. The user will need this password to change any Content Advisor settings, so the user might want to write it down for future reference.
- ii) View and adjust the ratings settings to reflect what the user thinks is appropriate content in each of four areas: language, adult, and violence.
- iii) Adjust what types of content other people can view, with or without the user's permission. The user can override content settings on a case-by-case basis.
- iv) Set up a list of Web sites that other people can never view, regardless of how the sites' contents are rated.
- v) Set up a list of Web sites that other people can always view, regardless of how the sites' contents are rated.
- vi) View and change the ratings systems and bureaus the user use.

Content Advisor is set to the most conservative (least likely to offend) settings when the user first turns it on. The user can adjust these settings to match the user's own preferences.

Not all Internet content is rated. If the user chooses to allow other people to view unrated sites on their computer, some of those sites could contain inappropriate material.

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2. THREATS IN NETWORKS

2.1. Network Security Issues

Networks have security problems for the following reasons:

i) ***Sharing.***

Because of the resource and workload sharing of networks, more users have potential to access networked systems than single computers. Perhaps worse, access is afforded to *more systems*, so that access controls for single systems may be inadequate in networks.

ii) ***Complexity of systems.***

We saw that an operating system is a complicated piece of software. Reliable security is difficult if not impossible on a large operating system, especially one not designed specially for security. A network has two or more possibly dissimilar operating systems. Therefore, a network operating/control system is likely to be more complex than an operating system for a single computing system. This complexity limits confidence in the security of network.

iii) ***Unknown perimeter.***

The expandability of a network also implies uncertainty about the network boundary. One host may be a node on two different works, so that resources on one network are accessible to the users of the other network as well. Although wide accessibility is an advantage, this unknown or uncontrolled group of possibly malicious users is a security disadvantage. A similar problem occurs when new hosts can be added to the network. Every network node must be able to react to the possible presence of new, untrusted hosts. See Figure 11

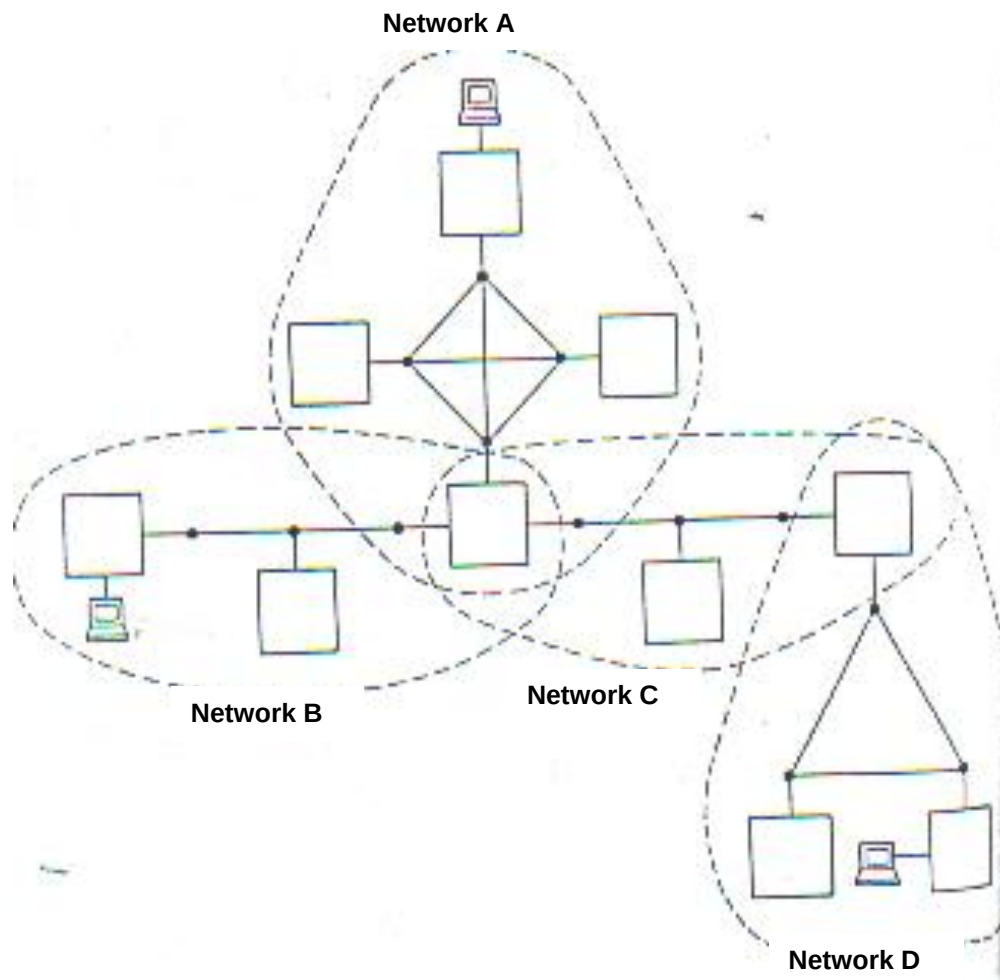


Fig. 11: Unclear Network Boundaries

iv) ***Many points of attack.***

A simple computing system is a self-contained unit. Access controls on one machine preserve the secrecy of data on that processor. However, when a file is stored in a network host remote from the user, the file may pass through many hosts to get to the user. Although the administrator of one host may enforce rigorous security policies, that administrator has no control over other hosts in the network. The user has to depend on the access control mechanisms of all of these systems.

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v) **Anonymity.**

An attacker can mount an attack from thousands of miles away and thus never have to touch the system attacked or come into contact with any of its administrators or users. The attack can be passed through many other hosts, in an effort to disguise from where the attack originated. Finally, computer-to-computer authentication is not the same as for humans to computers. Secure distributed authentication requires thought and attention to detail.

2.2. Security Threat Analysis

Let us recall how we performed security threat analysis in other situations. We looked at all the parts of a system, considered damage to confidentiality, integrity, availability and hypothesized the kinds of attacks that could cause this damage.

So, in a network we have:

- i) *local nodes* connected via,
- ii) *local communications link* to a
- iii) *local area network*, which also has
- iv) *local data storage*
- v) *local processes*, and
- vi) *local devices*.

The local area network is also connected to a

- i) *network gateway*, which gives access via
- ii) *network communications links* to
- iii) *network control resources*,
- iv) *network routers*, and
- v) *network resources*, such as data bases.

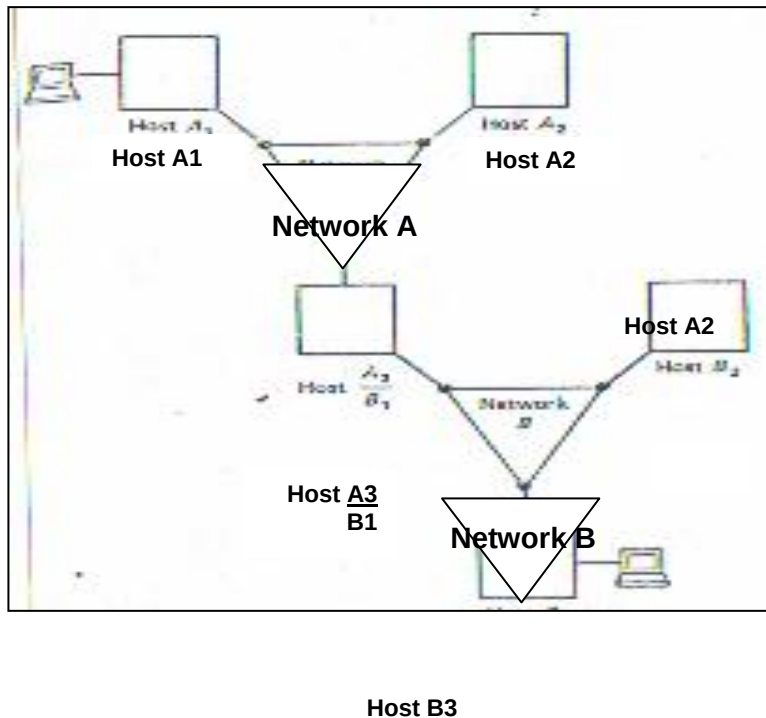


Fig. 12: Message Routing in a Network

For example, in Fig. 13, user A at workstation A wants to communicate with user B at workstation B, to use resources on the LAN to which B is attached, or use resources from LAN C elsewhere in the network. All of those functional needs are typical for users of networks. But let us now add a malicious outsider, M at workstation M, as in Fig. 13, and consider the negative effects M can cause.

- i) **Read communications** from A to B. These are exposed inside A's machine, at all places through the network, and inside B's machine. Thus a confidentiality attack can be mounted from practically any place in the network.
- ii) **Modify communications** from A to B. Again, these are exposed at all places through the network.
- iii) **Forge communications** allegedly from A to B. This is even easier than modifying a communication, because a forgery can be inserted any place in the network and does not require catching a communication in transit to modify. Because A does not deliver his communications personally, and because B might even never have met A. B has little basis for judging whether a communication purportedly from A is authentic.

- iv) **Inhibit communications** from A to B. Here again, M can achieve this result by invading A's machine, B's machine, routers between them, or communications links. M can also disrupt communications in general by flooding the network or any unique path on the network.
- v) **Inhibit all communications** passing through a point. If the point represents a unique path to or from a node, all traffic to or from that node is blocked. If the point is not unique, blocking it will shift traffic to other nodes, perhaps overburdening them.
- vi) **Read data at C or D.** M can impersonate A (who is authorized to access data at C or D). B might question a message out of character for Andy, but machine C and D will just apply the access controls for A. Alternatively, M can invade (run a program at) machines C or D to override access controls. Finally, M can search the network for machines that have weak or improperly administered access controls.

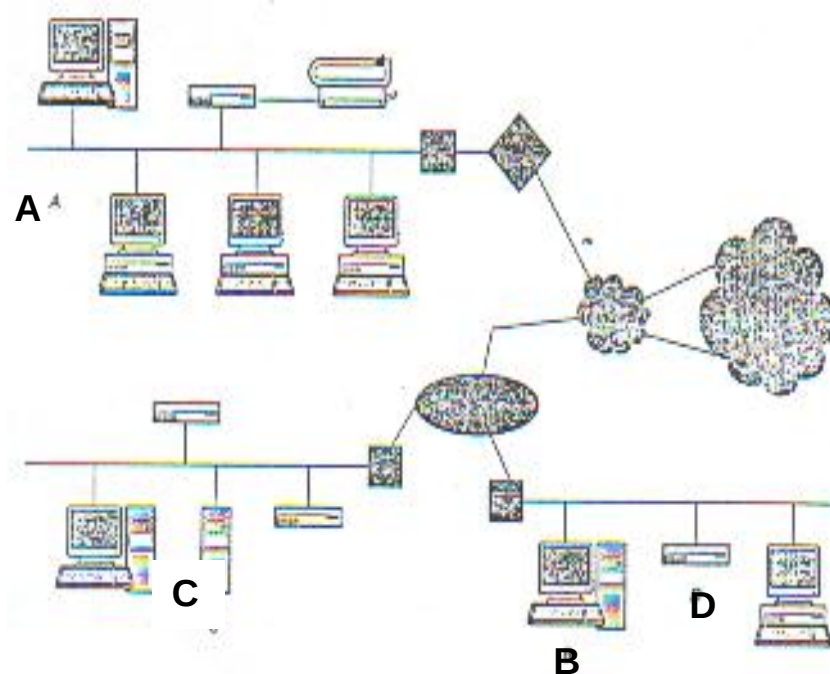


Fig. 13: Typical Network

- vii) **Modify or destroy data at C or D.** Here again M can impersonate A and do anything A could, or M can try to circumvent controls.

Thus, in summary, the threats are

- a) interception of data in transit
- b) access to programs or data at remote hosts
- c) modification of programs or data at remote hosts
- d) modification of data in transit
- e) insertion of communications impersonating a user
- f) insertion of a repeat of a previous communication
- g) blocking of selected traffic
- h) blocking of all traffic
- i) running a program at a remote host

How are all of these threats accomplished? We can group the potential threats into the following categories:

- a) wiretapping
- b) impersonation
- c) message confidentiality violations
- d) message integrity violations
- e) hacking
- f) code integrity violations
- g) denial of service

3. Network Security Controls

i) **Encryption**

Encryption is a very powerful tool for providing privacy, authenticity, integrity, and limited access to data. Because of the greater risks involved, networks often secure data with encryption, perhaps in combination with other controls.

In network applications, encryption can be applied either between hosts or between two applications. We consider these two forms of encryption below. Key distribution is always a problem with encryption. Encryption keys must be delivered to the sender and receiver in a secure manner.

ii) ***Access Control***

Encryption is especially good for protecting data within a network. However, access to data, programs and other resources of the network is also a serious concern in network security. In a single computing system, the number of authorized users may be limited because of the physical characteristics of the system. Often all users are within the same room or building, or are affiliated with the same organization. However, when a computing system is part of a network, users and even administrators may not know which other users may be connected to the same network. Thus, in a network environment, access control must protect a single system of the network and also prevent unauthorized users from passing through one system of a network to access other systems of the network. Two aspects of network access control that we consider now are protecting access points and authenticating network nodes.

iii) ***Authentication in Distributed Systems***

Application is not always completed on the user's local computer. Several computers joined on a network can share files, applications, and other resources. In this case, a user on one host may be unaware of the fact that a computation uses resources or facilities of some other hosts. Authentication and access control can all be done at the computer-to-computer level. There are then two problems to be solved:

- a) How can one host be assured of the authenticity of a remote host? Before sending sensitive data to a remote site, the host should be sure of the computer at the destination.
- b) How can a host be assured of the authenticity of a user on a remote host? Conversely, how can the user be assured of the authenticity of the remote host?

iv) ***Traffic Control***

We now turn to controls for a fairly sophisticated threat: traffic flow analysis. Few attackers can, or WAN to, monitor and analyze enough traffic to obtain mesh information from this form of attack. Furthermore, there are fairly simple defenses

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against such analysis.

Since an interceptor can tap all blocks of messages passing through the network the interceptor can determine who is communicating frequently with whom. Knowing that many messages are being transmitted between the personnel departmental and company management, a user might be able to infer that some substantial re-organization is about to occur. During wartime, the military has trouble planning surprise offensive because a large volume of messages to a particular place may arouse the suspicion of the enemy. In both these cases, it is not just the content, but also the mere existence of messages that is sensitive. In both cases, the attack is called a **traffic analysis**. The standard control is introduction of many spurious messages between points of low traffic. In this way, communication between the two sensitive places will not seem outstandingly heavy. Of course, this added traffic places an extra burden on the network, and service to all users may be degraded.

v) ***Privacy Enhanced Electronic Mail***

Privacy enhanced electronic mail addresses the specific problem of confidentiality, integrity, and authenticity in electronic mail (e-mail). It uses a combination of encryption, protocols, and data integrity controls to protect e-mail.

vi) ***Firewall***

A firewall is a process that filters all traffic between a protected or “inside” network and a less trustworthy or “outside” network. The purpose of a firewall is to keep “bad” things outside a protected environment. Firewalls implement a security policy. The policy might be to prevent any access from outside (while still allowing traffic to pass from the inside to the outside). Alternatively, it might be to permit accesses only from certain places, from certain users, or for certain activities. Part of the challenge of protecting a network with a firewall is determining the security policy that meets the needs of the installation.

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SOALAN / QUESTIONS:

1. List five (5) causes of network security problems.

2. How to identify *Access Control*?

3. What is a firewall?

RUJUKAN / REFERENCE:

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2. Crowder, David (1999), Getting on the Internet. IDG Books.
3. Windows Help and Support. XP Professional.